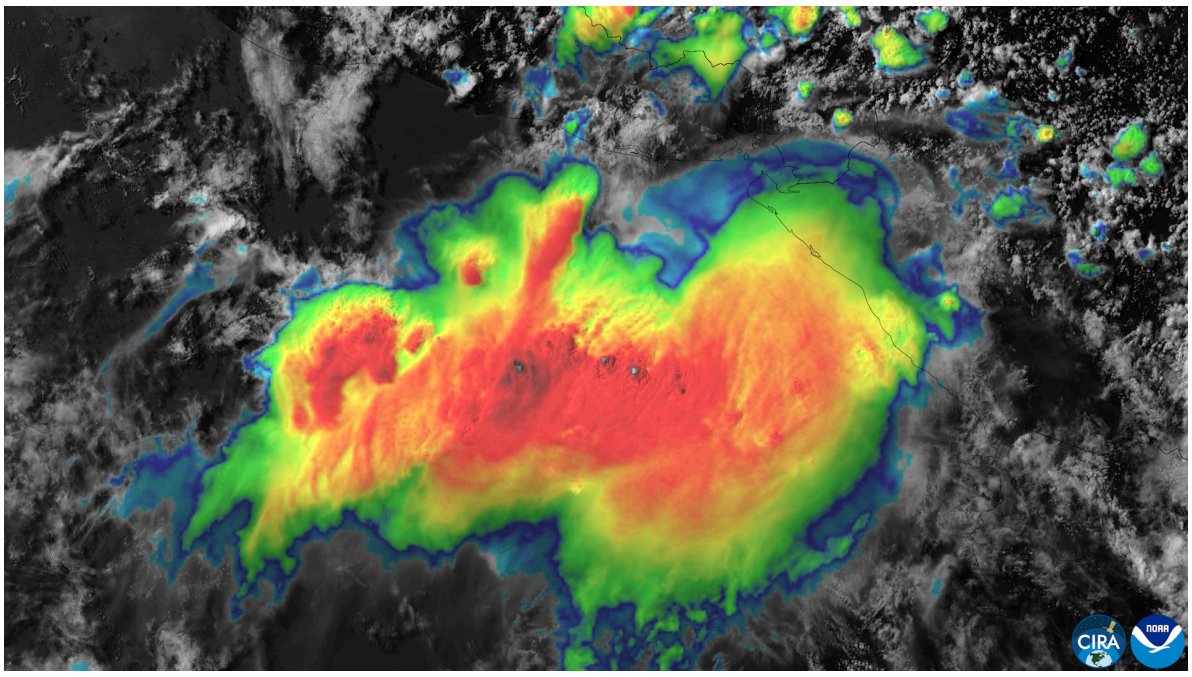
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT

TROPICAL STORM CRISTINA

(EP032026)

8-10 June 2026

John P. Cangialosi
National Hurricane Center
14 August 2026



GOES-19 SANDWICH SATELLITE IMAGE OF TROPICAL STORM CRISTINA AT 2100 UTC 9 JUNE. IMAGE COURTESY OF NOAA/CIRA.

Cristina was a short-lived tropical storm that brought coastal flooding, heavy rainfall, and minor damage to portions of Central America.

Tropical Storm Cristina

8-10 June 2026

SYNOPTIC HISTORY

Cristina appears to have originated from a disturbance embedded within the monsoon trough over the far eastern North Pacific. The system was first identified on 4 June just offshore of Central America, where it remained nearly stationary during the next several days. On 7 June, deep convection associated with the disturbance became increasingly organized off the coast of Costa Rica. By 0600 UTC 8 June, satellite imagery indicated that a well-defined low-level circulation had formed, marking the genesis of a tropical cyclone about 90 n mi west of the coast of southwestern Nicaragua. Based on a blend of satellite data, the initial intensity at the time of genesis is estimated at 35 kt. The “best track” chart of Cristina’s path is given in Fig. 1, with the wind and pressure histories shown in Figs. 2 and 3, respectively. The best track positions and intensities are listed in Table 1[1].

Cristina strengthened quickly after genesis, reaching a peak intensity of 45 kt just 6 h later while centered about 90 n mi west-northwest of Managua, Nicaragua. During this time, the cyclone moved northward within the low- to mid-level monsoonal flow. Shortly after attaining peak intensity, strong northerly vertical wind shear increased. These unfavorable winds, along with Cristina’s proximity to the rugged terrain of Central America, caused the deep convection to become increasingly disorganized by the evening of 8 June, and the cyclone began to weaken. On 9 June, the storm slowed and meandered just offshore of the Pacific coasts of Honduras and far northern Nicaragua. Although Cristina never made landfall, bands of heavy rainfall spread across coastal portions of El Salvador, Honduras, and Nicaragua throughout the day.

A slow westward motion commenced late on 9 June and continued the next day as a low- to mid-level ridge built to the north of the system. Although the wind shear had lessened, the circulation began to elongate, and by 0000 UTC 10 June, satellite images indicate that Cristina had opened into a trough of low pressure about 60 n mi south of the coast of El Salvador. The remnant trough dissipated about a day later just offshore of El Salvador.

METEOROLOGICAL STATISTICS

Observations in Cristina (Figs. 2 and 3) include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the

[1] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year’s storms are located in the *btk* directory, while previous years’ data are located in the *archive* directory.

Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Cristina.

Selected rainfall totals from land stations are given in Table 2.

Winds and Pressure

Cristina's maximum estimated wind speed of 45 kt at 1200 and 1800 UTC 8 June is based on a pair of ASCAT passes between those times that showed peak winds between 40 and 45 kt. Tropical storm conditions, especially in gusts, likely occurred along the immediate Pacific coast in portions of northwestern Nicaragua on 8–9 June.

The estimated minimum central pressure of 1002 mb is based on the Knaff-Zehr-Courtney pressure-wind relationship.

Cristina's genesis location was the fourth farthest east since reliable records began around 1970 in the eastern North Pacific basin. Francesca (1970), Miriam (1988), and Alma (2008) are the three tropical cyclones that formed farther east in the basin since 1970.

Rainfall and Flooding

Heavy rainfall spread across portions of Nicaragua, Honduras, and El Salvador on 9–10 June. Rainfall reports indicate that widespread totals of 3–5 inches fell along the Pacific coasts of those countries, however, there were isolated reports of up to 7 inches in El Salvador.

Coastal Flooding

The combination of Cristina's winds and southwesterly swell produced coastal flooding along portions of the Pacific coasts of Nicaragua, Honduras, and El Salvador. Coastal impacts included damage to roads, beach erosion, and damage to coastal properties in all three countries, including an amusement park in El Salvador.

CASUALTY AND DAMAGE STATISTICS

There were no reported casualties associated with Cristina. Flooding impacts along portions of the Pacific coasts of Nicaragua, Honduras, and El Salvador were generally minor, and no monetary estimates of the damage are currently available.

In Nicaragua, street flooding was reported in numerous coastal communities across the northwestern portion of the country. The flooding damaged roads, inundated homes, and

caused power outages. The island of Ometepe in Lake Nicaragua experienced a complete power outage during the event.

In El Salvador, heavy rainfall, accompanied by gusty winds, uprooted trees, damaged more than 100 homes and over 40 schools, and caused localized flooding. The worst conditions occurred near Santa Ana, where floodwaters prompted several rescues.

FORECAST AND WARNING CRITIQUE

Genesis

The genesis of Cristina was forecast reasonably well (Table 3). The disturbance that eventually became Cristina was first introduced in the Tropical Weather Outlook 186 h before genesis. The 7-day probability of formation was increased to the medium and high categories with lead times of 150 h and 84 h, respectively. However, the development probabilities were lowered back into the medium category after numerical model guidance became less supportive of tropical cyclone formation. The genesis potential was increased back to the high category 36 h prior to Cristina's formation. The 2-day formation probabilities also provided good lead time, with the low, medium, and high categories introduced 84 h, 54 h, and 36 h before genesis, respectively. In terms of location, Cristina formed near the eastern edge of most of the genesis regions highlighted in the Graphical TWO (Fig. 4).

Track

A verification of NHC official track forecasts for Cristina is provided in Table 4a. Official track forecast errors were lower than the previous 5-yr mean official errors at 12 and 24 h and were very close to the mean values at 36 and 48 h, although the latter statistics are based on a small number of verifying forecasts. A homogeneous comparison of the official track forecasts with selected guidance models is provided in Table 4b. NHC track forecasts outperformed all of the guidance at 12 and 24 h, while several models had slightly lower errors at 36 and 48 h. Although the official forecasts often indicated that Cristina would make landfall in Central America, post-storm analysis revealed that the cyclone dissipated just offshore.

Intensity

A verification of NHC official intensity forecasts for Cristina is provided in Table 5a. Official intensity forecast errors were lower than the previous 5-yr mean official errors from 12 to 36 h but exceeded the mean at 48 h, although the latter statistic is based on only two verifying forecasts. NHC correctly anticipated that Cristina would not strengthen significantly due to strong vertical wind shear and its close proximity to land. A homogeneous comparison of the official intensity forecasts with selected guidance models is provided in Table 5b. Several of the guidance models produced slightly lower forecast errors than the official forecasts.

Wind Watches and Warnings

Coastal wind watches and warnings associated with Cristina are given in Table 6.

ACKNOWLEDGEMENTS

Data and discussions provided by the meteorological services of Nicaragua, Honduras, and El Salvador were helpful in the preparation of this report.

Table 1. Best track for Tropical Storm Cristina, 8–10 June 2026.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
08 / 0600	10.8	87.4	1005	35	tropical storm
08 / 1200	11.4	87.6	1002	45	"
08 / 1800	11.9	87.7	1002	45	"
09 / 0000	12.4	87.7	1003	40	"
09 / 0600	12.6	87.5	1005	35	"
09 / 1200	12.5	87.5	1006	35	"
09 / 1800	12.6	87.7	1006	35	"
10 / 0000	12.5	88.0	1006	35	"
10 / 0600	12.4	88.3	1006	35	"
10 / 1200	12.3	88.6	1006	35	"
10 / 1800	12.4	88.8	1007	30	tropical depression
11 / 0000					dissipated
08 / 1200	11.4	87.6	1002	45	maximum wind and minimum pressure

Table 2. Selected rainfall totals for Tropical Storm Cristina, 8–10 June 2026.

Location	Total Rain (in)
Honduras	
Sabana Grande *(13.81N 87.26W)*	3.38
San Pedro Sula *(15.51N 88.07W)*	3.13
Jesus De Ortoro *(14.49N 87.98W)*	2.51
Eapo *(15.27N 88.54W)*	3.13
El Chiabo *(14.13N 87.13W*	2.97
El Salvador	
Telica	5.34
La Paz Centro	5.77
Dirianda	6.89
El Rosa Rio	5.20

Table 3. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	84	186
Medium (40%-60%)	54	150
High (>60%)	36	84 (36)

The number in parentheses indicates when the disturbance was re-introduced into the high category in the TWO.

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Cristina, 8–10 June 2026. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**19.6**	**31.3**	41.8	50.7				
OCD5	29.2	72.8	135.0	186.4				
Forecasts	8	6	4	2				
OFCL (2021-25)	21.0	31.9	41.7	50.4	60.8	73.3	99.6	124.1
OCD5 (2021-25)	35.9	72.5	113.0	154.2	196.4	239.0	325.0	405.8

Table 4b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Cristina, 8–10 June 2026. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	19.6	28.8	47.7	67.8				
OCD5	29.2	83.6	148.4	214.1				
GFSI	34.9	55.8	49.0	**43.0**				
EMXI	24.8	29.6	**44.0**	**64.4**				
CMCI	21.0	30.5	**42.4**	**59.2**				
HWFI	29.4	54.7	90.3	128.9				
HFAI	46.1	99.2	129.3	189.9				
HFBI	44.8	89.0	128.7	193.6				
HCCA	22.5	34.3	**25.9**	**34.8**				
GFEX	23.5	31.8	**27.0**	**29.9**				
TVCE	24.1	32.8	**29.5**	**58.9**				
Forecasts	8	4	3	1				

Table 5a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Cristina, 8–10 June 2026. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.1**	**6.7**	**8.8**	17.5				
OCD5	8.1	8.5	5.5	2.5				
Forecasts	8	6	4	2				
OFCL (2021-25)	5.6	8.8	10.9	12.7	13.7	14.0	15.5	16.7
OCD5 (2021-25)	7.6	12.6	16.4	19.4	21.1	21.2	21.0	21.9

Table 5b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Cristina, 8–10 June 2026. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 5a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.1	6.0	8.3	20.0				
OCD5	8.1	**2.8**	**4.3**	**2.0**				
DSHP	**2.8**	**3.8**	**2.0**	**8.0**				
LGEM	3.6	**4.4**	**3.0**	**9.0**				
HWFI	4.8	7.0	9.0	**12.0**				
HFAI	5.2	**4.8**	**5.7**	**7.0**				
HFBI	4.6	**4.8**	8.3	**6.0**				
GFSI	4.8	**5.8**	15.3	21.0				
EMXI	3.1	**3.6**	**4.0**	**2.0**				
HCCA	4.0	**4.2**	**5.7**	**2.0**				
IVCN	**2.9**	**3.8**	**2.3**	**9.0**				
Forecasts	8	5	3	1				

Table 6. Watch and warning summary for Tropical Storm Cristina, 8–10 June 2026.

Date/Time (UTC)	Action	Location
08 / 1500	Tropical Storm Warning issued	Entire coast of El Salvador
08 / 1500	Tropical Storm Warning issued	Nicaragua Pacific coast from Puerto Sandino to Honduras border
08 / 1800	Tropical Storm Warning issued	Honduras Pacific coast
10 / 1500	Tropical Storm Warning discontinued	Nicaragua Pacific coast
10 / 2100	Tropical Storm Warning discontinued	Honduras Pacific coast
10 / 2100	Tropical Storm Warning changed to a Tropical Storm Watch	Entire coast of El Salvador
11 / 0600	Tropical Storm Watch discontinued	Entire coast of El Salvador

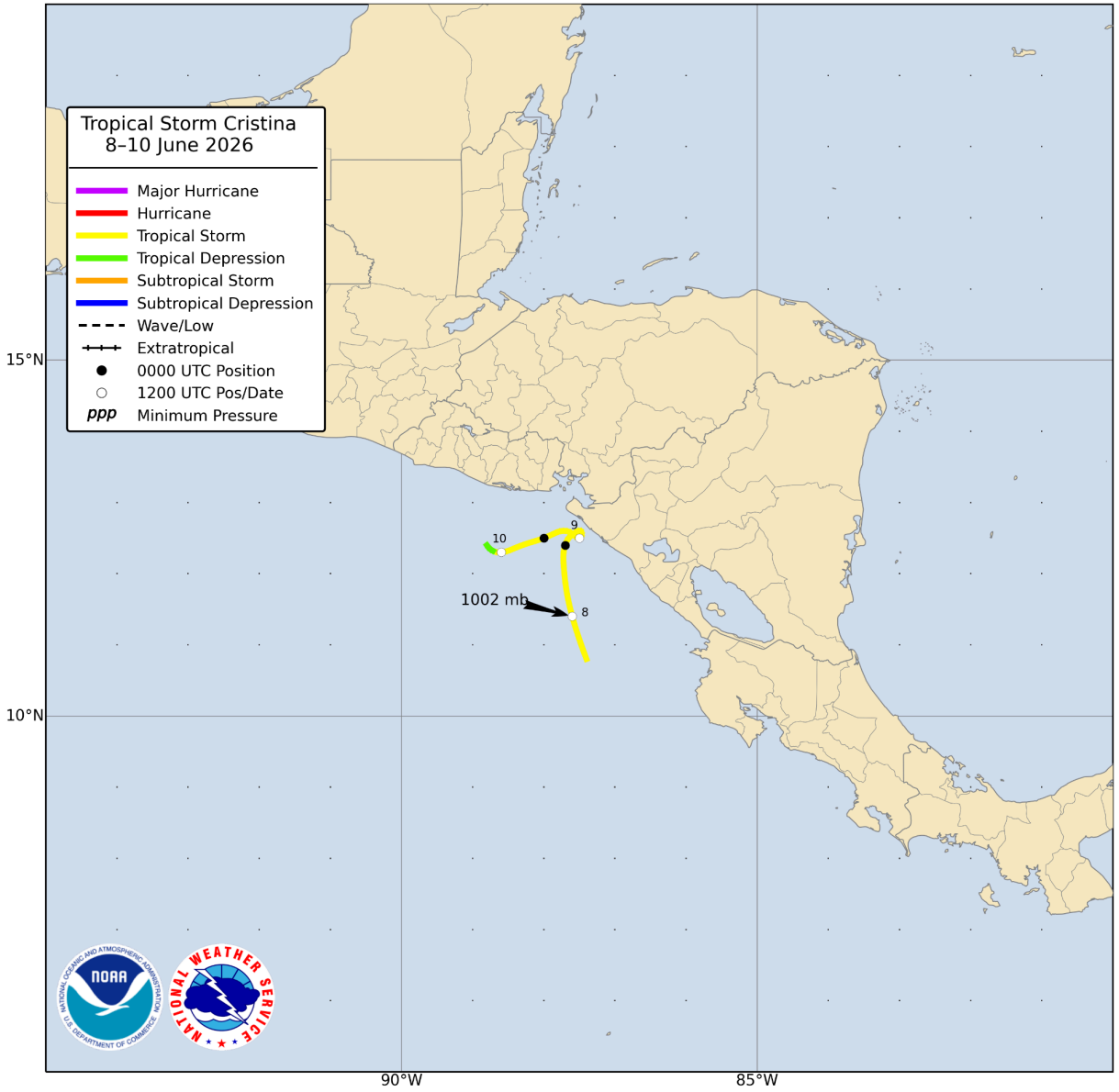


Figure 1. Best track positions for Tropical Storm Cristina, 8-10 June 2026.

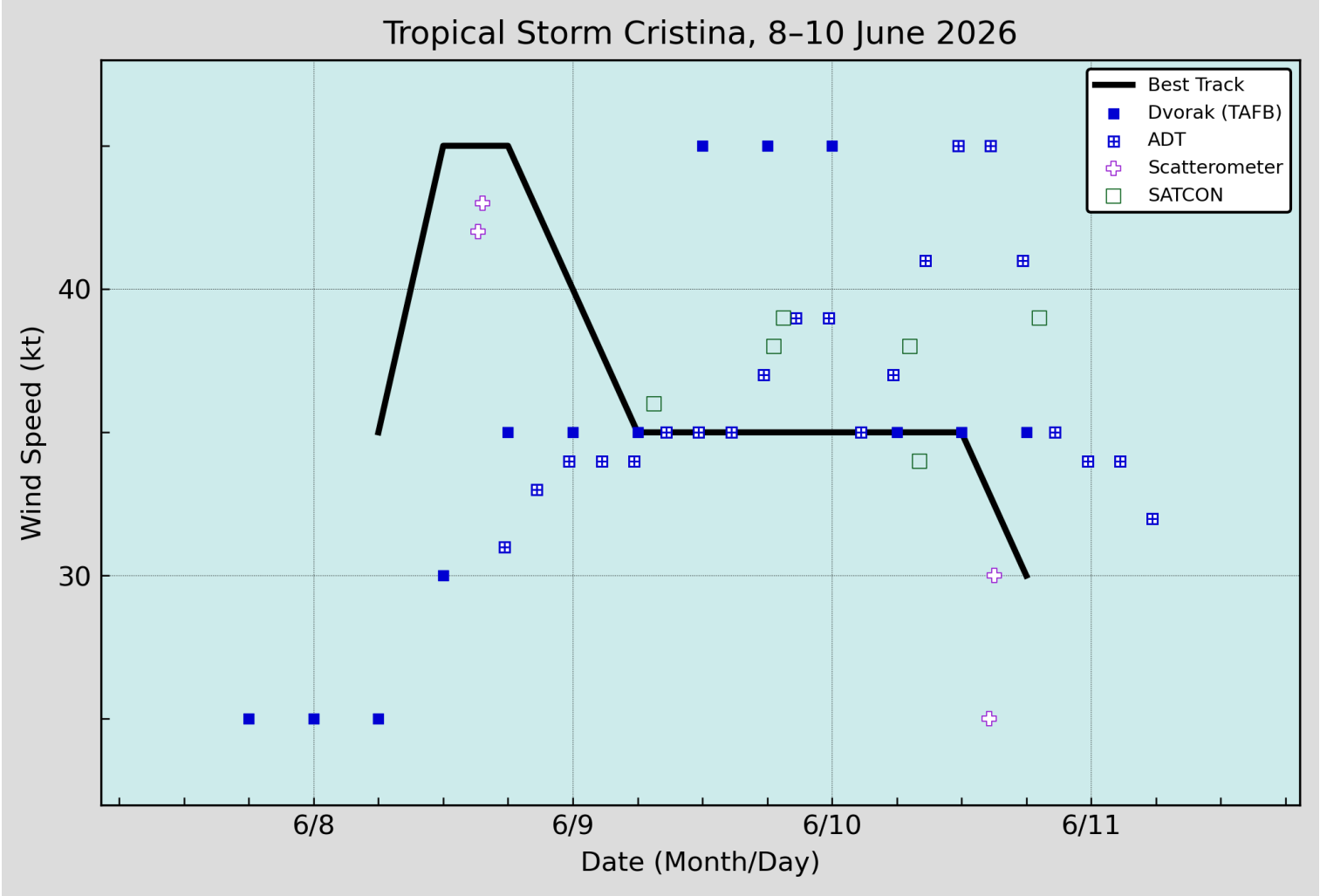


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Cristina, 8-10 June 2026. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

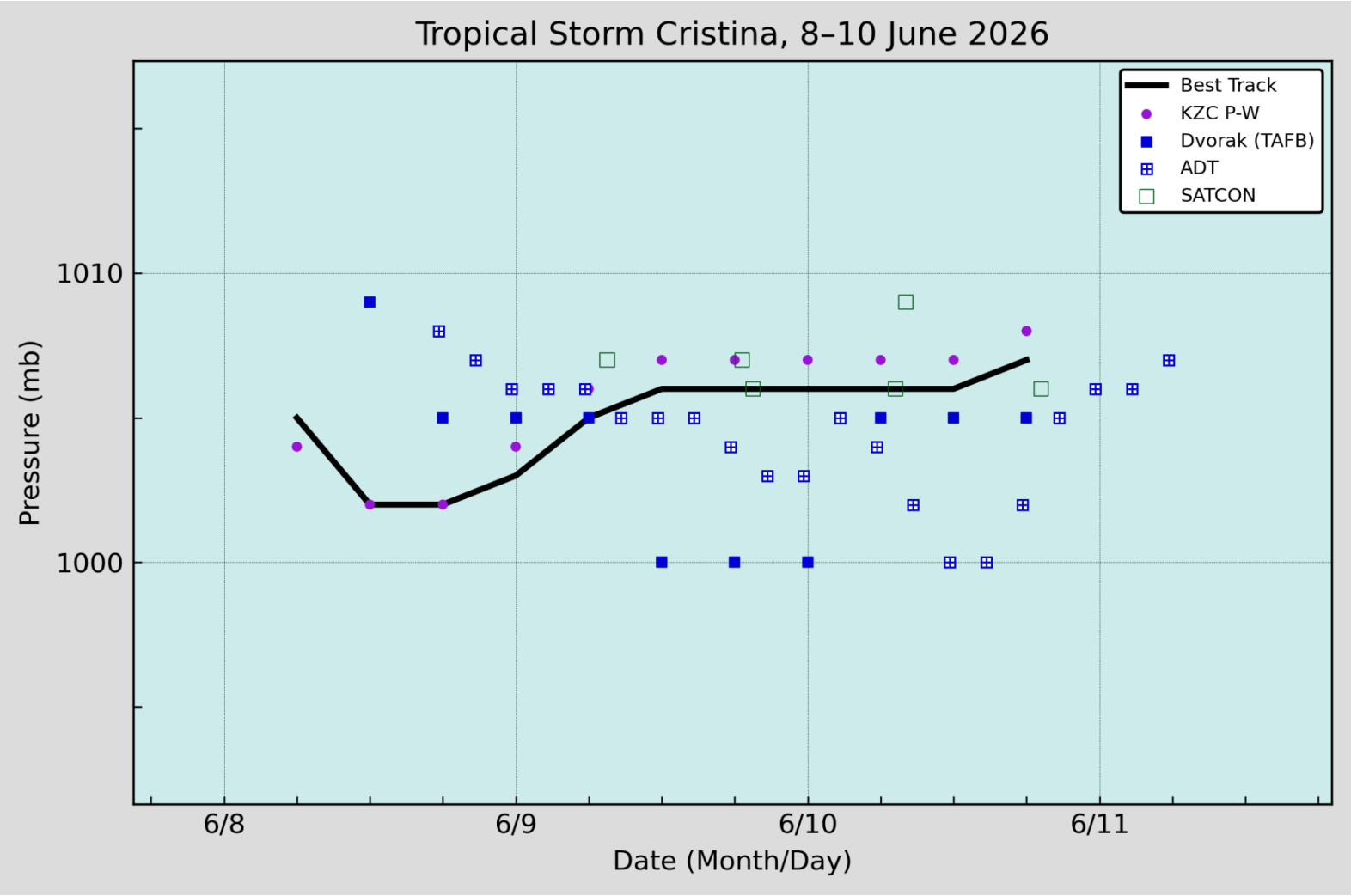


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Cristina, 8-10 June 2026. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

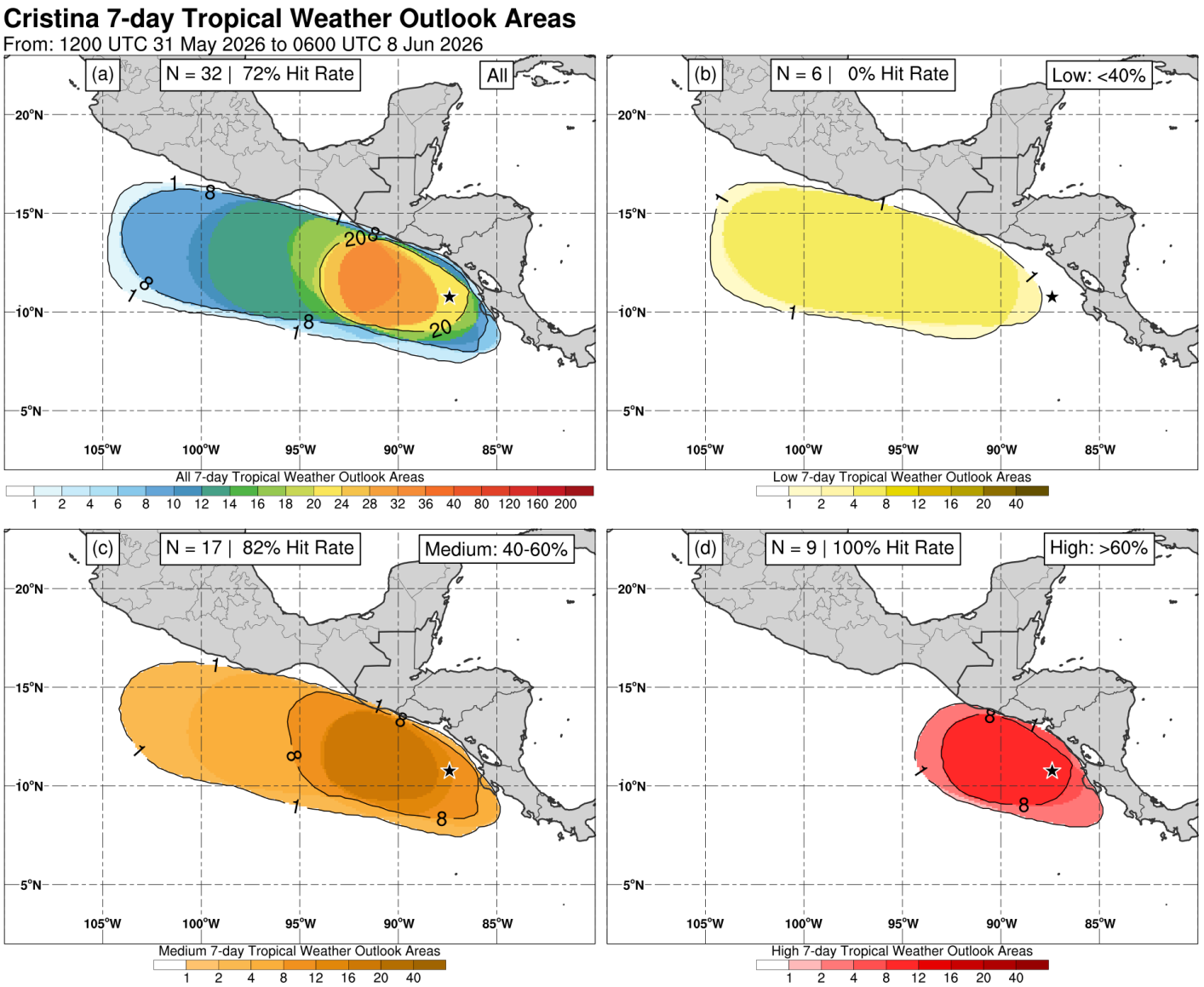


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Tropical Storm Cristina for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.